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Department of Computational Finance

Analyzing Fund Flow Patterns and Market Efficiency in Pakistan with Predictive Models

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UNITED NATIONS SUSTAINABLE DEVELOPMENT GOALS

EXECUTIVE SUMMARY

This project, *"Analyzing Fund Flow Patterns and Market Efficiency in Pakistan with Predictive Models,"* aims to explore the relationship between mutual fund flows and market efficiency in the Pakistani financial market. Despite the growing importance of mutual funds and index investing, there is limited research on how investor sentiment and fund flows impact market dynamics in Pakistan. This project addresses this gap by leveraging data science and machine learning techniques to predict fund flows and develop data-driven investment strategies.

The study begins with the collection of historical index and mutual fund data from the Pakistan Stock Exchange (PSX). The data will be cleaned, preprocessed, and visualized to understand trends, patterns, and correlations. Statistical analysis, including correlation and regression, will be conducted to identify relationships between fund flows and market efficiency indicators. Machine learning models will then be implemented to forecast fund flows, enabling the team to determine optimal weight allocation for funds in an index.

The project is expected to provide insights into investor behavior and contribute to more efficient portfolio management practices within Pakistan's financial market. In alignment with the United Nations Sustainable Development Goals, the project promotes *Decent Work and Economic Growth* and *Industry, Innovation, and Infrastructure* by applying analytical and predictive techniques to support informed financial decision-making.

The project spans from August 2025 to June 2026 and is undertaken by a team of four students under the supervision of Ms. Ubaida Fatima, Assistant Professor in the Department of Mathematics. The outcomes will not only fill a research gap but also offer practical applications for investors and policymakers in the Pakistani financial market.

CHAPTER: 01

INTRODUCTION

1.1 OVERVIEW OF MUTUAL FUNDS

Mutual funds are collective investment vehicles that pool money from multiple investors to invest in a diversified portfolio of securities such as stocks, bonds, and other financial instruments. These funds are managed by professional fund managers who make investment decisions on behalf of the investors, aiming to achieve specific financial goals. Mutual funds provide investors with access to a diversified portfolio, which helps in spreading risk and reducing the impact of individual security performance on overall returns.

A mutual fund works on the principle of ownership through units or shares, where each investor owns units proportional to their investment in the fund. The Net Asset Value (NAV) of a mutual fund represents the per-unit value of its assets, which fluctuates based on the performance of the underlying securities. Investors can earn returns through capital appreciation, dividends, or interest income generated by the fund’s portfolio.

Mutual funds are regulated investment products, and in Pakistan, the Securities and Exchange Commission of Pakistan (SECP) oversees their operations. Fund management companies, such as conventional or Islamic asset management firms, offer a variety of mutual funds tailored to different risk appetites and investment objectives. Common types of mutual funds include equity funds, debt funds, balanced funds, and money market funds. Equity funds primarily invest in stocks, debt funds invest in bonds and fixed-income securities, balanced funds combine equities and fixed-income assets, and money market funds focus on short-term, low-risk investments.

Investors in mutual funds are broadly categorized into retail investors—individuals investing for personal financial goals—and institutional investors, such as banks, insurance companies, pension funds, and other large financial organizations. Mutual funds can also be open-ended, allowing investors to enter or exit the fund at any time, or closed-ended, where units are fixed for a specific period and traded on stock exchanges.

The performance of a mutual fund depends on factors such as the skill of the fund manager, the market conditions, economic indicators, and investor sentiment. Diversification, risk management, and proper asset allocation are key strategies employed by fund managers to achieve consistent returns.

Overall, mutual funds play a critical role in the financial market by providing an accessible investment option for individuals and institutions, promoting capital formation, and supporting efficient allocation of financial resources across the economy.

1.2 PROBLEM IDENTIFICATION

The Pakistani financial market has witnessed significant growth over the past decades, with increasing participation in equity markets and mutual funds. However, despite this growth, research on index and mutual funds remains limited, particularly in understanding the dynamics of investor sentiment, fund flows, and their impact on market efficiency. Most existing studies focus on developed economies, where market structures, investor behavior, and regulatory frameworks differ substantially from those in Pakistan.

This research gap presents challenges for investors and policymakers. Without a clear understanding of how fund flows interact with market efficiency in the local context, it becomes difficult to make informed investment decisions or develop policies that support market stability and growth. Investor sentiment, which can drive short-term market movements, and fund flows, which reflect the collective behavior of investors, are critical indicators that remain underexplored in Pakistan.

Addressing this gap is crucial for multiple reasons. First, it can provide investors with insights into optimal fund allocation strategies and potential market trends. Second, it can assist policymakers in identifying patterns that may influence market efficiency, liquidity, and stability. Finally, the study contributes to the broader field of finance and data science by applying predictive models and machine learning techniques to a market that has been largely overlooked in empirical research.

By analyzing fund flow patterns and their relationship with market efficiency, this project aims to bridge the knowledge gap, offering practical and data-driven insights for both investors and regulatory authorities in Pakistan.

1.3 SCOPE

This project focuses on the Pakistani financial market, which is considered an under-developed economy with relatively primitive financial infrastructure compared to more developed countries. Despite this, a variety of complex financial products exist, offering diverse techniques to address investment and risk management challenges. In Pakistan, the Securities and Exchange Commission of Pakistan (SECP) and the State Bank of Pakistan (SBP) act as regulatory authorities overseeing financial institutions and market practices.

Currently, financial institutions use established methods, such as historical analysis, to monitor and evaluate fund flows, portfolio performance, and risk measures. However, relying on a single approach may limit the accuracy and reliability of these assessments. This project aims to apply multiple analytical and predictive techniques—including statistical analysis, correlation studies, and machine learning models—to examine fund flow patterns and their impact on market efficiency. By exploring different modeling approaches, we can better understand trends, make more accurate predictions, and provide actionable insights for investment decision-making.

1.4 OBJECTIVE OF THIS STUDY

Machine learning techniques are widely used in the finance field for analyzing and predicting fund flows, which are closely related to investment strategies and market efficiency.

This study helps to utilize machine learning models to predict fund flows in the context of the Pakistani financial market.

By examining historical fund data, the research aims to forecast future investment strategies and determine the optimal weight allocation for each fund in an index.

Also, this study will provide insights into the relationship between fund flows and market efficiency, which can help investors, fund managers, and policymakers make more informed decisions for portfolio management and market performance.

CHAPTER: 02

LITERATURE REVIEW

Yamani, Ehab (2023) investigates the informational role of fund flows in predicting mutual fund performance for profitable investment strategies. The study uses a sample of 2,217 US equity mutual funds from 2000 to 2018, analyzing past fund flows and returns. The research builds predictive models to forecast whether a fund's next-period return will be positive or negative, using past fund flow direction as the main input. Lasso Regression (L1-Regularized Regression) and Random Forests (RF) are employed for prediction. The study finds that fund flows can successfully predict future returns, and implementing strategies based on these predictions leads to outperformance. The study is limited to US data and includes a restricted set of predictors, suggesting the need for testing in other markets or with additional features.

Larsson, Erik & Wergeland, Jacob (2020) explore the relationship between index fund flows and market efficiency in the S&P 500, analyzing whether changes in market efficiency cause changes in index fund flows. The study uses data from 633 index funds and S&P returns from 2000 to 2019. Methods include calculating the Hurst exponent via RS and DFA, fund flow calculations, and Granger causality tests. Results indicate that lower market efficiency reduces index fund flows, suggesting that market efficiency drives changes in index fund investment. Limitations include the focus on US data and reliance primarily on the Hurst exponent, highlighting the need for broader studies in other markets using multiple variables or models.

Barber, B. M., Huang, X., & Odean, T. (2016) examine which performance factors mutual fund investors consider when deciding to invest, and how this differs between sophisticated and less sophisticated investors. The study uses U.S. actively managed equity mutual funds from the CRSP database, including returns, flows, and exposures to market beta, size, value, momentum, and industry factors. Econometric regressions link fund flows to decomposed returns into alpha and factor returns, with proxies for investor sophistication. The findings reveal that investors focus mainly on market beta returns, often confusing factor returns with skill. Sophisticated investors react more accurately, while unsophisticated investors misinterpret factor returns. The study is limited to U.S. equity mutual funds and measures investor sophistication indirectly, without assessing long-term effects on fund success.

Jadoon, A. K., Mahmood, T., Sarwar, A., Javaid, M. F., & Iqbal, M. (2024) aim to predict the movement of the KSE 100 Index using machine learning, considering economic, social, and political factors. The study uses monthly KSE 100 data and applies an Artificial Neural Network (ANN) with a Backpropagation algorithm. A Long Short-Term Memory (LSTM) network is employed to forecast index movements. The model demonstrates high prediction accuracy, confirming that deep learning algorithms like LSTM are effective for forecasting complex, volatile financial markets. Future research is suggested to incorporate additional financial features and advanced deep learning models.

Yaqoob, A., & Abdullah, S. M. (2025) develop an LSTM-based deep learning model to predict closing prices of ten major stocks in the Pakistani market across different sectors using PSX historical OHLCV data. The model is trained on multi-year data and evaluated using R-squared (R^2). Results indicate strong predictive performance ($R^2 > 0.87$) for stable, high-activity sectors, while performance is weaker for high-risk, low-liquidity stocks. The study confirms the applicability of LSTM networks in emerging markets and highlights the need for further improvements to handle volatile stocks and external shocks.

Hassan, H., Niaz, A., Qureshi, J. A., & Rooh, S. (2025) investigate the effect of South Asian stock market fluctuations on the KSE-100 Index. Monthly KSE-100 pricing data is analyzed using Multiple Regression and ARIMA models to examine inter-market linkages. The study finds that past volatility and lagged conditional variance strongly predict future volatility. Hybrid models, including SVM-GARCH and Neural Networks, outperform simpler GARCH models. The findings help investors diversify risk and assist policymakers in understanding regional market impacts.

Hassan, H., Niaz, A., Qureshi, J. A., & Rooh, S. (2025) also study the impact of macroeconomic factors on KSE-100 performance, using monthly data from July 2014 to June 2024. Multiple Regression Analysis and ARIMA models are employed to assess the influence of exchange rates, FDI, and Balance of Trade (BOT). Results indicate significant effects of exchange rate fluctuations, FDI, and trade surplus on stock market performance. The study provides actionable insights for policymakers, investors, and analysts in managing economic risks.

Idrees, M., Sial, M. H., & Hassan, N. U. (2025) compare four deep learning models for predicting KSE-100 daily closing prices, focusing on LSTM with Attention. Using daily data from 2008 to 2021 (3221 rows), models include ANN, RNN with Attention, LSTM with Attention, and GRU with Attention. The LSTM-Attention model achieves the highest accuracy (R^2: Training 0.9996, Validation 0.9980, Testing 0.9921). The study highlights the effectiveness of the attention mechanism for sequential financial data and proposes a robust architecture for stock price prediction.

CHAPTER: 03

METHADOLOGY

3.1 Conceptual Framework of Fund Flow Analysis

Fund flows refer to the net movement of money into or out of mutual funds or investment vehicles over a period of time. Analyzing fund flows is critical for understanding investor behavior and assessing market efficiency. Fund flows provide information about investor sentiment, revealing which assets or funds are attracting or losing investment. Positive net fund flows often indicate bullish sentiment, whereas negative flows reflect caution or pessimism.

The analysis of fund flows over time allows us to observe essential patterns, including trends, cyclic behavior, and short-term fluctuations. Trends indicate sustained increases or decreases in fund allocations over time, reflecting overall investor confidence or market perception. Cyclic patterns emerge due to periodic economic or regulatory factors affecting investment decisions, while short-term fluctuations capture temporary shifts in investor sentiment. Understanding these dynamics is essential for forecasting future fund movements and developing predictive investment strategies.

3.2 Data Collection and Preprocessing

The study collects historical data for mutual funds and indices listed on the Pakistan Stock Exchange (PSX), focusing on the KSE-30 and KSE-100 indices. The dataset spans the period from January 2020 to October 2025. Preprocessing was performed to ensure data accuracy and consistency. Columns with missing or zero values were filled using the forward-filling method, while rows where index weightage was zero were removed. Company symbols were standardized to create a unique identifier for each company, and non-essential columns were reduced to focus on relevant features. The data was normalized to make it suitable for machine learning algorithms, and visualization techniques were applied to detect trends, patterns, and relationships within the dataset.

Table 1: 3.2 Data Collection

DATE	SYMBOL	COMPANY	PRICE	IDX WT	FF BASED SHARES	FF BASED MCAP	ORD SHARES	ORD SHARES MCAP	VOLUME
12/23/2024	AIRLINK	Air Link Communication Limited	215.53	0.87	98817308	21298094393	98817308	21298094393	10266206
12/24/2024	AIRLINK	Air Link Communication Limited	210.38	0.86	98817308	20789185257	98817308	20789185257	8765161
12/26/2024	AIRLINK	Air Link Communication Limited	213.91	0.89	98817308	21138010354	98817308	21138010354	3834991
12/27/2024	AIRLINK	Air Link Communication Limited	215.74	0.90	98817308	21318846028	98817308	21318846028	3848039
12/30/2024	AIRLINK	Air Link Communication Limited	217.38	0.87	98817308	21480906413	98817308	21480906413	1497575
12/31/2024	AIRLINK	Air Link Communication Limited	220	0.86	98817308	21739807760	98817308	21739807760	5090891
1/1/2025	AIRLINK	Air Link Communication Limited	222.43	0.86	98817308	21979933818	98817308	21979933818	4690463
1/2/2025	AIRLINK	Air Link Communication Limited	222.89	0.86	98817308	22025389780	98817308	22025389780	2703007
1/3/2025	AIRLINK	Air Link Communication Limited	217.98	0.84	98817308	21540196798	98817308	21540196798	3144956
1/6/2025	AIRLINK	Air Link Communication Limited	210.97	0.84	98817308	20847487469	98817308	20847487469	2301398
1/7/2025	AIRLINK	Air Link Communication Limited	209.55	0.83	98817308	20707166891	98817308	20707166891	1700776
1/8/2025	AIRLINK	Air Link Communication Limited	212.82	0.86	98817308	21030299489	98817308	21030299489	2438606
1/9/2025	AIRLINK	Air Link Communication Limited	196.65	0.81	98817308	19432423618	98817308	19432423618	2565186
1/10/2025	AIRLINK	Air Link Communication Limited	189.64	0.77	98817308	18739714289	98817308	18739714289	15954589
1/13/2025	AIRLINK	Air Link Communication Limited	197.55	0.80	98817308	19521359195	98817308	19521359195	4214945
1/14/2025	AIRLINK	Air Link Communication Limited	200.29	0.76	98817308	19792118619	98817308	19792118619	3219264
1/15/2025	AIRLINK	Air Link Communication Limited	205.81	0.79	98817308	20337590159	98817308	20337590159	8437242
1/16/2025	AIRLINK	Air Link Communication Limited	200.9	0.77	98817308	19852397177	98817308	19852397177	2526028
1/17/2025	AIRLINK	Air Link Communication Limited	204.45	0.93	118580769	24243838222	118580769	24243838222	2814720
1/20/2025	AIRLINK	Air Link Communication Limited	201.24	0.91	118580769	23863193954	118580769	23863193954	3425103
1/21/2025	AIRLINK	Air Link Communication Limited	197.97	0.90	118580769	23475434839	118580769	23475434839	4171463
1/22/2025	AIRLINK	Air Link Communication Limited	194.83	0.90	118580769	23103091224	118580769	23103091224	2392566

kse30_daily_data

Data of KSE-30 Index from PSX.

3.3 FEATURE ENGINEERING

3.3.1 Daily Returns

Daily log returns for each company were calculated to capture the percentage change in price over consecutive trading days. This helps in understanding short-term price movements and forming the basis for volatility calculations.

$$\text{Daily\_Return}_t = \ln\left(\frac{P_t}{P_{t-1}}\right)$$

3.3.2 Moving Averages

Short-term and long-term trends were identified by calculating moving averages of price and volume. The 7-day and 30-day rolling averages for price (MA7_Price, MA30_Price) and 7-day rolling average for volume (MA7_Volume) were computed to smooth out short-term fluctuations.

7-day Moving Average (MA7_Price):

$$\text{MA7\_Price}_t = \frac{1}{7}\sum_{i=0}^{6} P_{t-i}$$

30-day Moving Average (MA30_Price):

$$\text{MA30\_Price}_t = \frac{1}{30}\sum_{i=0}^{29} P_{t-i}$$

7-day Moving Average of Volume (MA7_Volume):

$$\text{MA7\_Volume}_t = \frac{1}{7}\sum_{i=0}^{6} V_{t-i}$$

3.3.3 Volatility

Volatility was estimated as the 30-day rolling standard deviation of daily returns. This feature captures the magnitude of price fluctuations and is essential for assessing risk in fund flow prediction.

$$\text{Volatility}_t = \sqrt{\frac{1}{29}\sum_{i=0}^{29}\left(\text{Daily\_Return}_{t-i} - \overline{\text{Daily\_Return}}\right)^2}$$

3.3.4 Momentum Features

Momentum indicators were generated using percentage changes in price over 5-day and 10-day periods (Momentum_5d, Momentum_10d) to capture short-term price trends and market sentiment.

5-day Momentum:

$$\text{Momentum\_5d}_t = \frac{P_t - P_{t-5}}{P_{t-5}}$$

10-day Momentum:

$$\text{Momentum\_10d}_t = \frac{P_t - P_{t-10}}{P_{t-10}}$$

3.3.5 Relative Strength Index (RSI)

RSI was calculated over a 14-day window to quantify overbought or oversold conditions in stock prices, helping in identifying potential turning points in price movements.

$$\text{RSI}_t = 100 - \frac{100}{1 + RS_t}$$

$$RS_t = \frac{\text{Average Gain over 14 days}}{\text{Average Loss over 14 days}}$$

3.3.6 Bollinger Bands

Upper and lower Bollinger Bands were computed using a 20-day rolling standard deviation around the 30-day moving average of price. The band width (BB_Width) measures price dispersion and volatility in the market.

Upper Band:

$$BB_{\text{Upper},t} = MA30\_Price_t + 2 \cdot \sigma_{20,t}$$

Lower Band:

$$BB_{\text{Lower},t} = MA30\_Price_t - 2 \cdot \sigma_{20,t}$$

Band Width:

$$BB_{\text{Width},t} = \frac{BB_{\text{Upper},t} - BB_{\text{Lower},t}}{MA30\_Price_t}$$

3.3.7 Project-Specific Features

- **delta_wt, delta_price, delta_volume:** daily changes in index weightage, price, and volume were calculated to track short-term dynamics.

$$\Delta WT_t = WT_t - WT_{t-1}$$

$$\Delta P_t = P_t - P_{t-1}$$

$$\Delta V_t = V_t - V_{t-1}$$

- **prev_ff_based_mcap:** lagged fund-flow-based market capitalization was used to capture historical fund trends.

$$\text{PREV\_FF\_MCAP}_t = \text{FF\_MCAP}_{t-1}$$

- **flow_proxy:** daily change in fund-flow-based market capitalization normalized by the previous value, serving as a stable measure of fund movements.

$$\text{Flow\_Proxy}_t = \frac{\text{FF\_MCAP}_t - \text{FF\_MCAP}_{t-1}}{\text{FF\_MCAP}_{t-1}}$$

- **volume_price_ratio:** ratio of daily volume to price, representing interaction between trading activity and price levels.

$$\text{Volume\_Price\_Ratio}_t = \frac{V_t}{P_t}$$

CHAPTER: 05

RECOMMENDATIONS

REFERENCES